

Introduction to Data Science and Engineering

- R markdown



Some materials courtesy of
Rafael A. Irizarry, and are modified
from the original version.

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School of Computing and Data Science
University of Hong Kong

Slide deck originally created by RB Luo



In this lecture

- Markdown language & R markdown
- Markdown Basics



In this lecture

- **Markdown language & R markdown**
- Markdown Basics



R markdown for stylish texts, inline code and results

Assignments are handed-out and received as R markdown format, so please get familiar with it

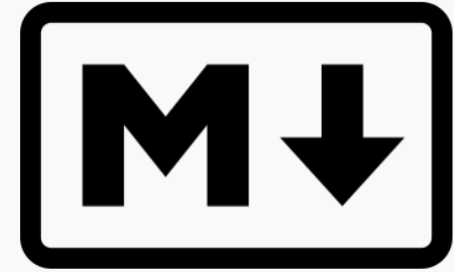
See also: <http://rafalab.dfci.harvard.edu/dsbook/reproducible-projects-with-rstudio-and-r-markdown.html#r-markdown>

Markdown language

- **Markdown: a lightweight markup language for creating formatted text using a plain-text editor.**
- Created in 2004 by John Gruber and Aaron Swartz as a markup language that is appealing to human readers in its source code form.
- **Widely used in blogging, instant messaging, online forums, collaborative software, documentation pages, and README files**
 - Github & Gitlab
 - Discord, Slack, Notion
 - Visual Studio Code
 - Moodle

See also: <https://en.wikipedia.org/wiki/Markdown>

Markdown



Internet media type	text/markdown ^[1]
Uniform Type Identifier (UTI)	net.daringfireball.markdown
Developed by	John Gruber and Aaron Swartz
Initial release	March 19, 2004 (17 years ago) ^{[2][3]}
Latest release	1.0.1 (December 17, 2004 (16 years ago) ^[4])
Type of format	Markup language
Extended to	pandoc, MultiMarkdown, Markdown Extra, CommonMark, ^[5] RMarkdown ^[6]
Open format?	Yes ^[7]
Website	daringfireball.net/projects/markdown/ 



Work with Markdown

- **Plenty of online resources:**

- <https://stackedit.io/>
- <https://markdownlivepreview.com/>

ChatGPT 5 ▾

↑ Share ...

List me some resources for learning and writing markdowns online, put them in a table please.

Here's a table of popular resources you can use to learn and practice writing Markdown online:

Resource	Type	Key Features	URL
Markdown Guide	Tutorial & Reference	Comprehensive syntax guide, extended syntax, cheat sheet	https://www.markdownguide.org ↗
GitHub Markdown Guide	Tutorial & Reference	GitHub Flavored Markdown (GFM) documentation, examples	https://docs.github.com/en/get-started/writing-on-github ↗
Dillinger	Online Editor	Live preview, export to HTML/PDF, cloud integration	https://dillinger.io ↗
StackEdit	Online Editor	WYSIWYG + Markdown hybrid, offline mode, Google Drive sync	https://stackedit.io ↗

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HackMD	Collaboration Tool	Real-time collaborative Markdown editing, diagrams, integrations	https://hackmd.io
Obsidian Help	App Documentation	Guides for using Markdown in Obsidian note-taking	https://help.obsidian.md
Markable	Online Editor	Instant HTML preview, export options, simple UI	https://markable.in
Learn Markdown in Y Minutes	Quick Reference	Concise syntax overview with examples	https://learnxinyminutes.com/docs/markdown

Do you want me to also **add a quick “Markdown syntax cheat sheet”** to go with this table so you can reference common formatting instantly?

↓

Copy

Ask anything

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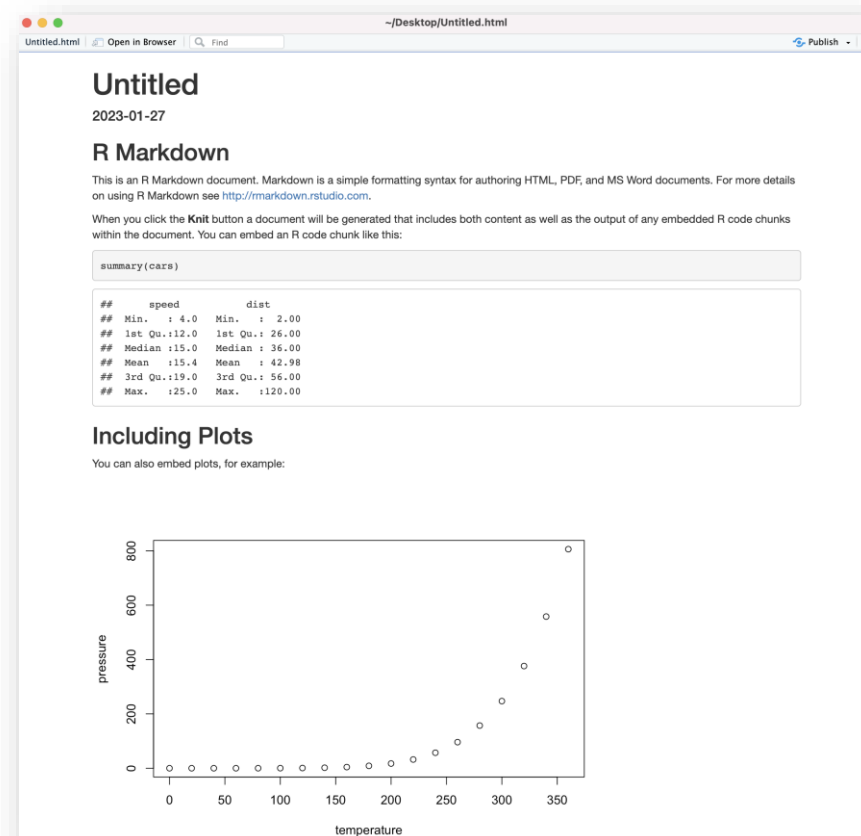
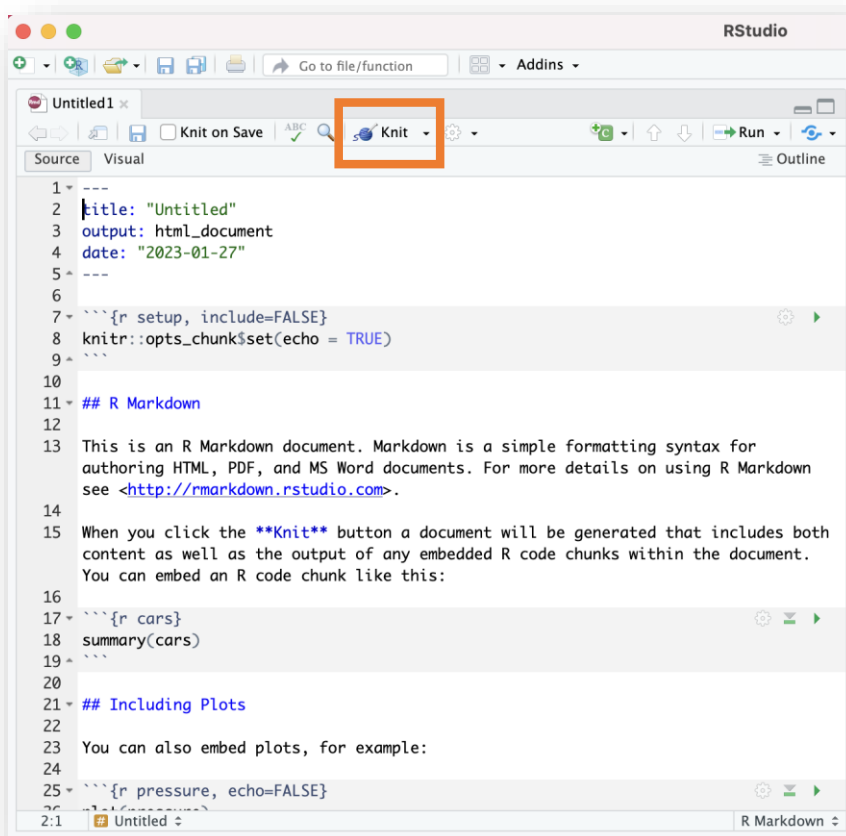
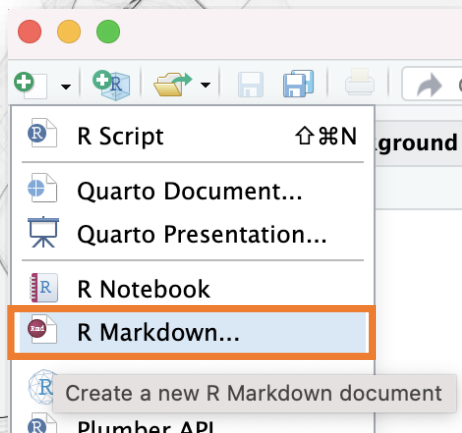
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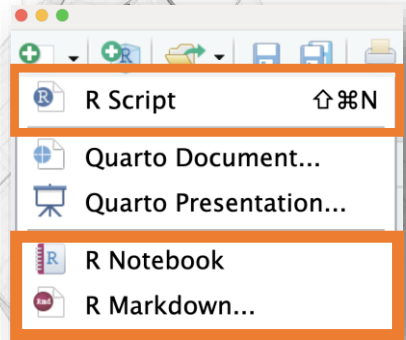
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Work with R Markdown



R Markdown / R Notebook

- **R Markdown (.Rmd):** for creating polished, reproducible documents like reports or tutorials.
 - Codes are executed when *knitted*
- **R Notebook:** an interactive extension of R Markdown, ideal for exploratory analysis, prototyping, and development.
 - Codes are executed in real-time with inline results.
- **R Scripts (.R):** plain text files used to write and execute R code, no additional formatting or documentation features.
 - Codes are run in an R console





In this lecture

- Markdown language & R markdown
- **Markdown Basics**

Download the example markdown

Download
Learn_data_science_with_R.Rmd
from Moodle

Learning Data Science with R

CS2501

2025-08-09

Learning Data Science with R

Welcome to the **ultimate guide** to learning **Data Science** with **R**. This guide is designed to help you master data science concepts while learning R, one of the most popular programming languages for data analysis and visualization.

Table of Contents

1. [Introduction](#)
2. [What is Data Science?](#)
3. [Why Learn R?](#)
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 - [Statistics and Probability](#)
 - [Statistical Models and Machine Learning](#)
5. [Resources](#)
6. [Conclusion](#)

Introduction

Data Science is a multidisciplinary field that combines **mathematics**, **statistics**, **computer science**, and **domain knowledge** to extract insights from data.

"R is not just a statistical language; it's a complete ecosystem for data science."

What is Data Science?

Data science involves five key steps:

1. **Data Collection**
2. **Data Cleaning**
3. **Exploratory Data Analysis (EDA)**
4. **Model Building**
5. **Data Visualization and Communication**

```

1 # Learning Data Science with R
2
3 Welcome to the ultimate guide to learning Data Science with R. This
  guide is designed to help you master data science concepts while learning R,
  one of the most popular programming languages for data analysis and
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5 ---
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16     - [Statistical Models and Machine Learning]
17       (#statistical-model-and-machine-learning)
17 6. [Resources](#resources)
18 7. [Conclusion](#conclusion)
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Introduction

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Titles

Use # for title, section title, subsection, etc.

```
8 # Learning Data Science with R
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```

```
81 ## Key Topics
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83 ### Programming with R
84
85 R is the primary language for this course. Key concepts include:
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87 - Data Structures: Vectors, matrices, lists, and data frames.
88 - Data Manipulation: Using `tidyverse` to filter, group, and summarize data.
89 - Writing Functions: Creating reusable code.
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Key Topics

Programming with R

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- **Data Structures**: Vectors, matrices, lists, and data frames.
- **Data Manipulation**: Using `tidyverse` to filter, group, and summarize data.
- **Writing Functions**: Creating reusable code.



Titles

Up to 7 levels of titles

- Enough for your homework?

```
# Header 1
## Header 2
### Header 3
#### Header 4
##### Header 5
##### Header 6
##### Header 7?
Text
```

Header 1

Header 2

Header 3

Header 4

Header 5

Header 6

Header 7?

Text

Split line / horizontal rule

Three or more hyphens (---), you can also use underscores (____), or asterisks(***)

```
8 # Learning Data Science with R
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24 5. [Our Progress](#our-progress)
25 6. [Resources](#resources)
26 7. [Conclusion](#conclusion)
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28 ---
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5. Our Progress
6. Resources
7. Conclusion



Line break

Create line break could be tricky.

```
## Linebreak myths
```

```
This is one line.
```

```
---
```

```
This is still  
one line.
```

```
---
```

```
But now we have  
two lines.
```

```
---
```

```
The safe way is to use  
a blank line in between paragraphs
```

```
---
```

```
Or you can use the <br> tag. <br> Like this.
```

```
---
```

Linebreak myths

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Line break

Two spaces instead of one will create a linebreak

Linebreak myths

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Like this.

Line break

Recommended ways: create a blank line, use html tag `<br>`

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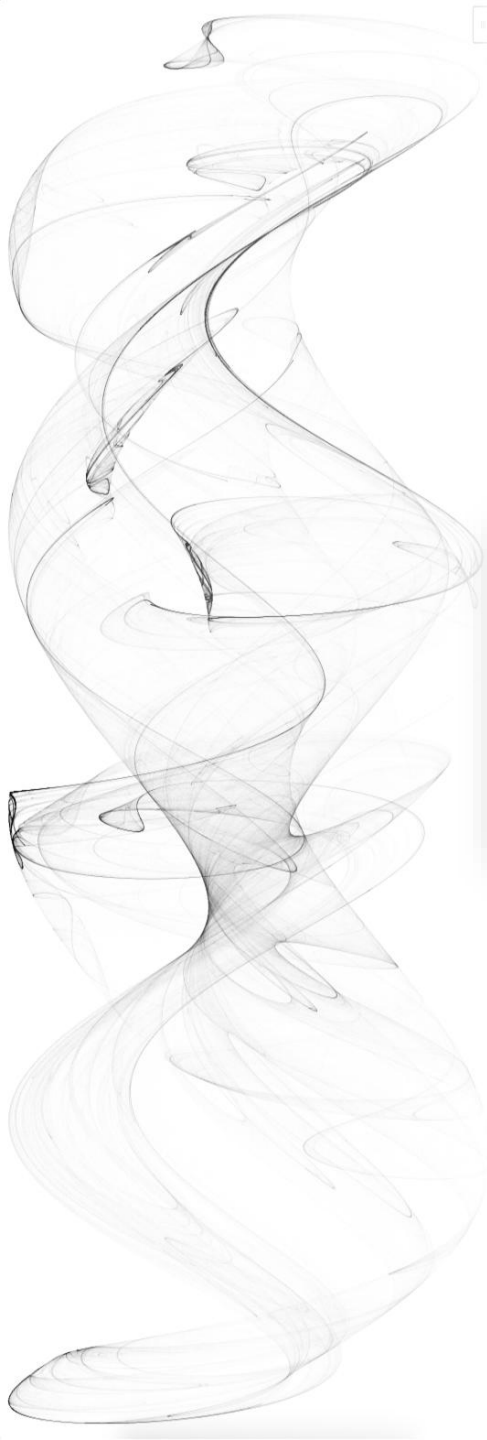
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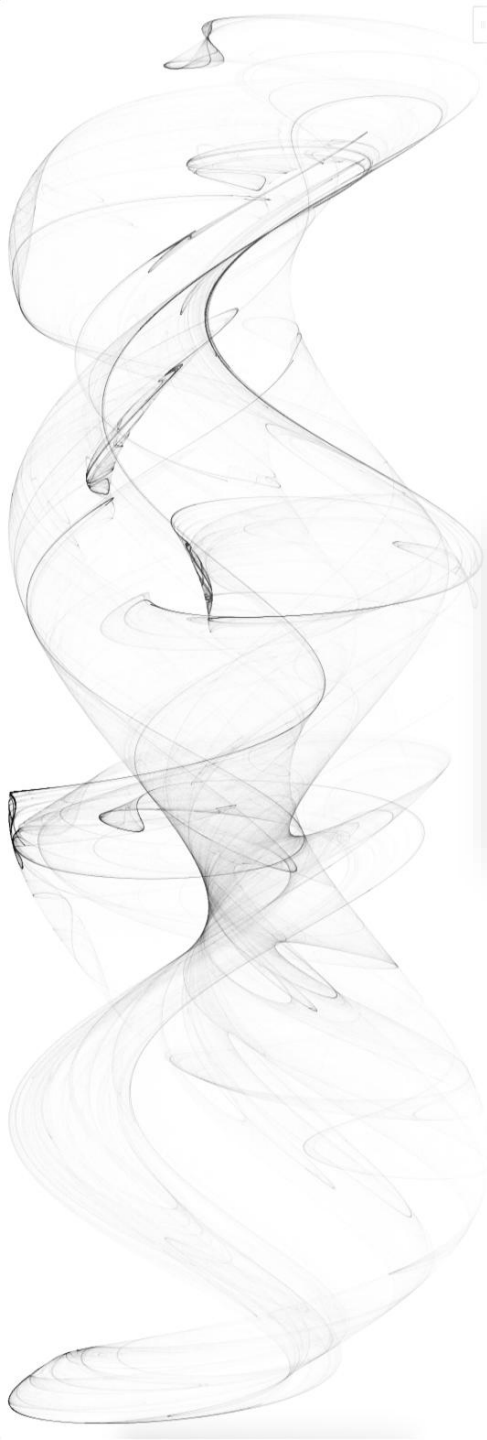
Italic by *

```
29 ▾ ## Introduction
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   **mathematics**, **statistics**, **computer science**, and
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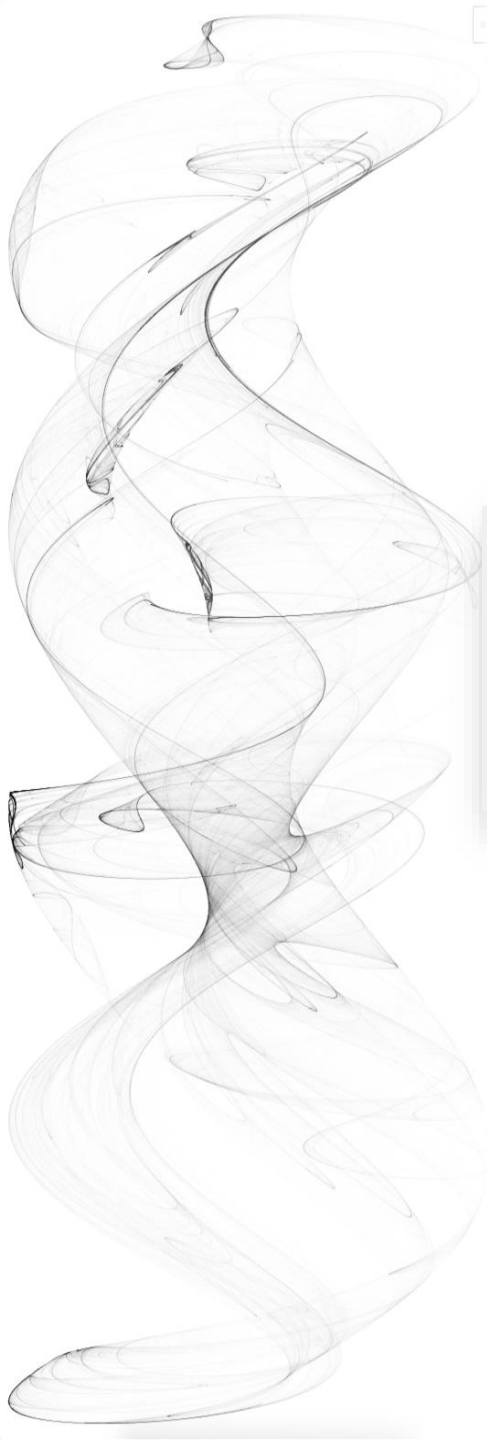
Bold by **

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Data Science is a multidisciplinary field that combines **mathematics, statistics, computer science, and domain knowledge** to extract insights from data.

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Bullet points

Dash, space, your contents: – CONTENT

```
49 - ## Why Learn R?  
50  
51 R is a **powerful language** for statistical analysis, data  
    visualization, and machine learning. Some benefits include:  
52  
53 - Excellent libraries for data science (e.g., `dplyr`,  
    `ggplot2`).  
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A decorative graphic on the left side of the slide consisting of multiple overlapping, wavy, grey lines that create a sense of motion and depth.

Numbered list

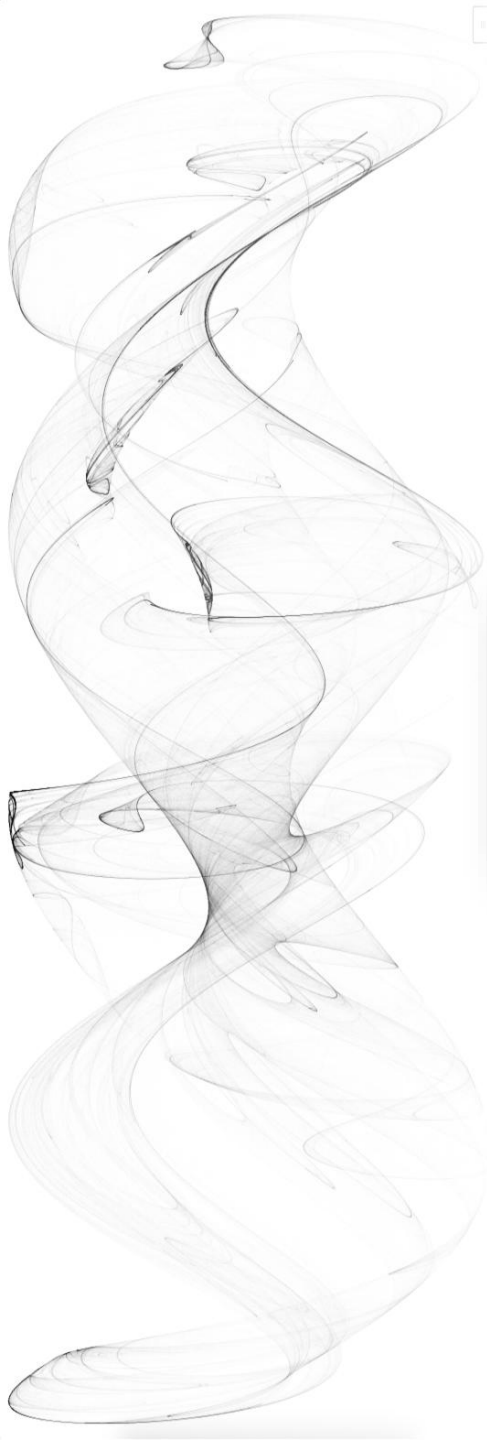
Number, dot, space, your contents: 1 . CONTENT

Numbered list

1. First item
2. Second item
4. No, this is still the third item
4. Now this is the fourth item

Numbered list

1. First item
2. Second item
3. No, this is still the third item
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Checklist

Dash, bracket, space, your contents: - [] CONTENT

```
192 ▾ ## Our Progress
193
194 - [x] Set up R and RStudio
195 - [x] Learn basic R syntax
196 - [ ] Learn data manipulation with `dplyr`
197 - [ ] Explore data visualization with `ggplot2`
198 - [ ] Build your data science project
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Our Progress

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- ☒ Learn basic R syntax
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Link

Syntax: [TEXT_TO_SHOW] (LINK)

Link could also be section titles



made with mematic

JANE-CLARK.TUMBLR

```
183 ▾ ## Resources
184
185 Here are some recommended resources to learn R and data science:
186
187 1. Books
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190 2. Cheat Sheets
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194   - \[Stack Overflow\](https://stackoverflow.com)
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Image

Syntax: ! [CAPTION] (IMAGE_LINK)

You can also leave the caption placeholder empty

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58 ![*R packages for data
science*](https://www.business-science.io/assets/069_r_logo_boar
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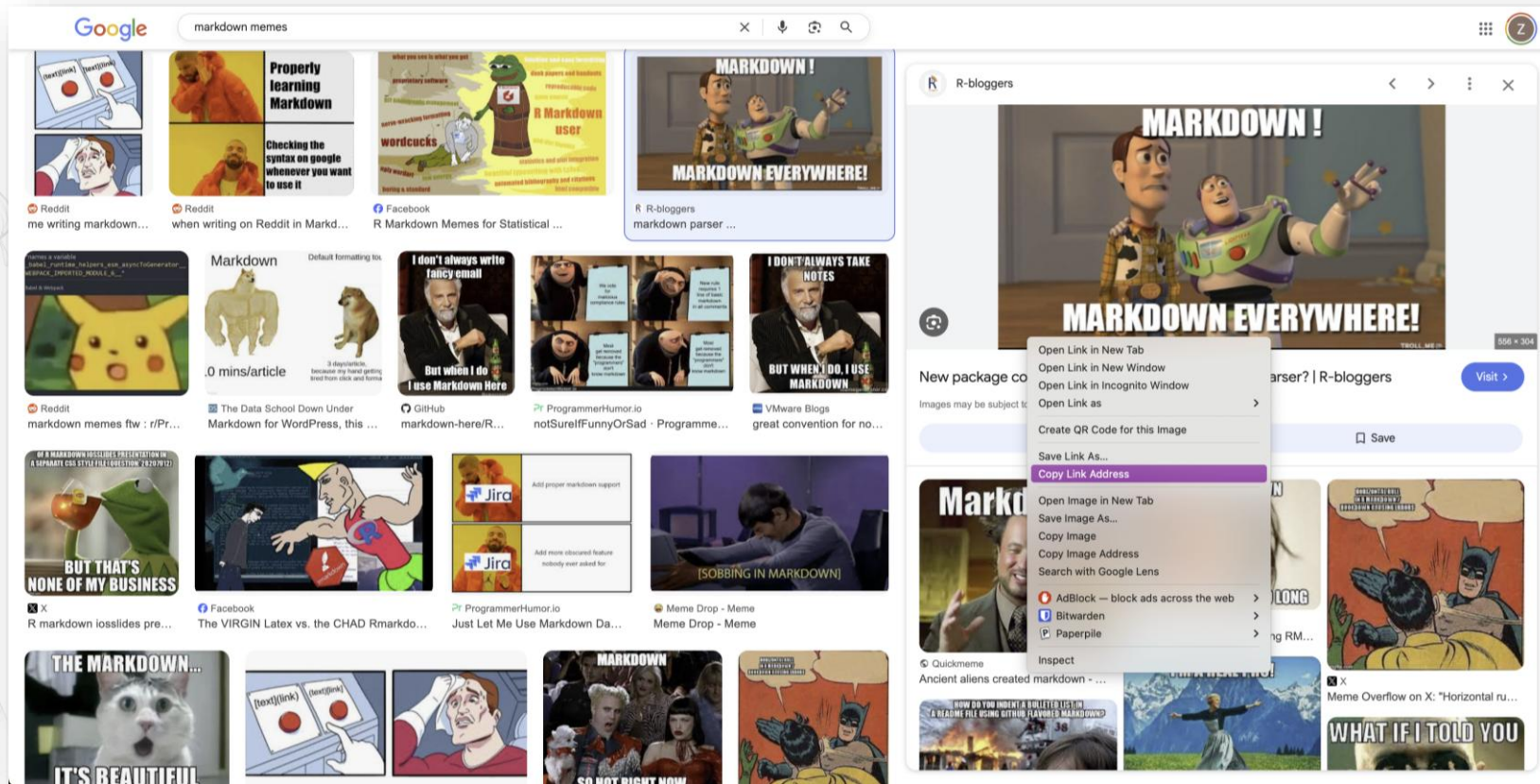
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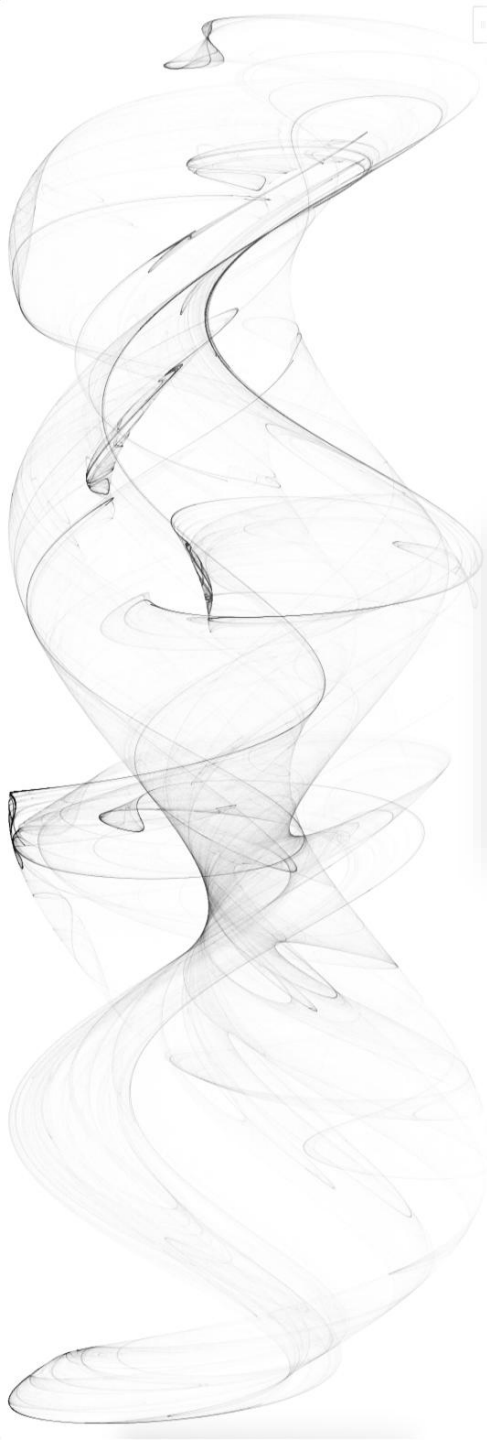
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R packages for data science

Copy the link of an image online





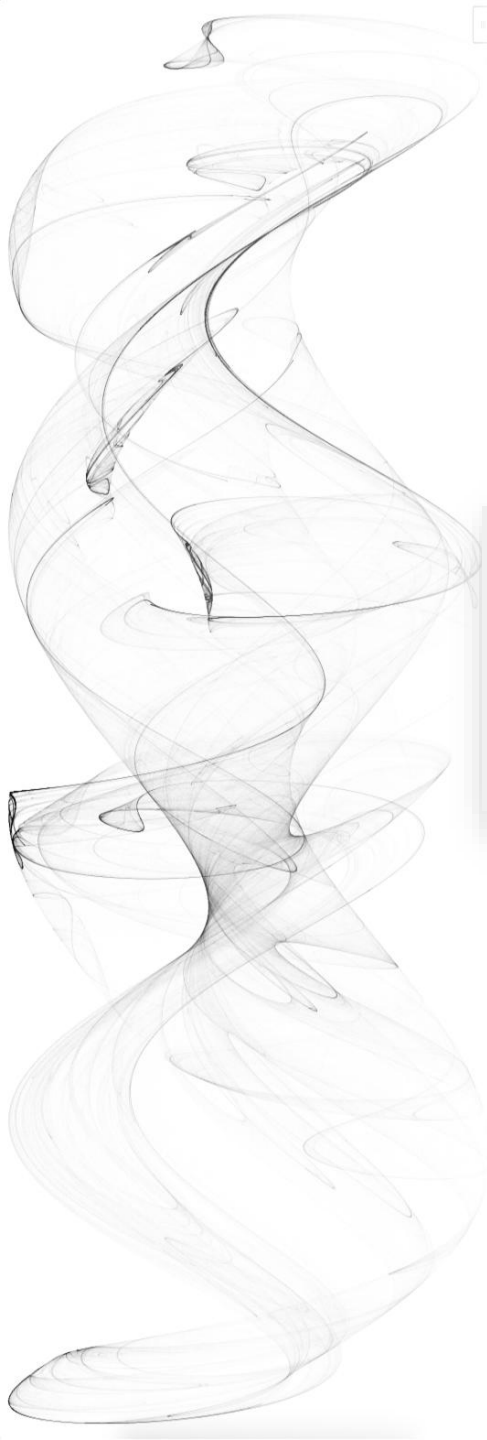
Quote by >

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Inline codes

Backquote `

Unfortunately, `Powerpoint` doesn't support this

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Code block

Wrap your codes in between two lines, each with three backquotes `` `

Programming with R

R is the primary language for this course. Key concepts include:

- **Data Structures**: Vectors, matrices, lists, and data frames.
- **Data Manipulation**: Using `tidyverse` to filter, group, and summarize data.
- **Writing Functions**: Creating reusable code.

Example:

```
```{r, echo=TRUE}  
Create a vector
numbers <- c(1, 2, 3, 4, 5)

Calculate the mean
mean(numbers)
```
```

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```

```
## [1] 3
```

Code block

The text immediately following the opening ``` in a code block specifies the programming language or syntax highlighting to be applied to the code within the block.

- Yes, R markdown can highlight other languages (e.g. Python).
- No, codes will not be run, and don't do this in your homework.

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```python
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numbers = np.arange(1, 6)
print(np.mean(numbers))
```
```

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```

Code block – R markdown specialties

Syntax:

```
```{r, <name>, <options>}  
Code goes here
```
```

- It is highly recommended to assign a name to each code block through <name>, for easy reference
- What goes to <options>:
 - echo=TRUE: if TRUE, source codes will be shown, otherwise hidden
 - warning=FALSE: if FALSE, warnings will be hidden

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```{r, echo=TRUE}  
Create a vector
numbers <- c(1, 2, 3, 4, 5)

Calculate the mean
mean(numbers)
```
```

Example:

```
```{r, echo=TRUE, warning=FALSE}  
Split data into training and testing sets
library(caret)
set.seed(123)
split <- createDataPartition(mtcars$mpg, p = 0.8, list = FALSE)
train <- mtcars[split,]
test <- mtcars[-split,]

Train a linear regression model
model <- train(mpg ~ ., data = train, method = "lm")

Summarize the model
summary(model)
```
```

Tables

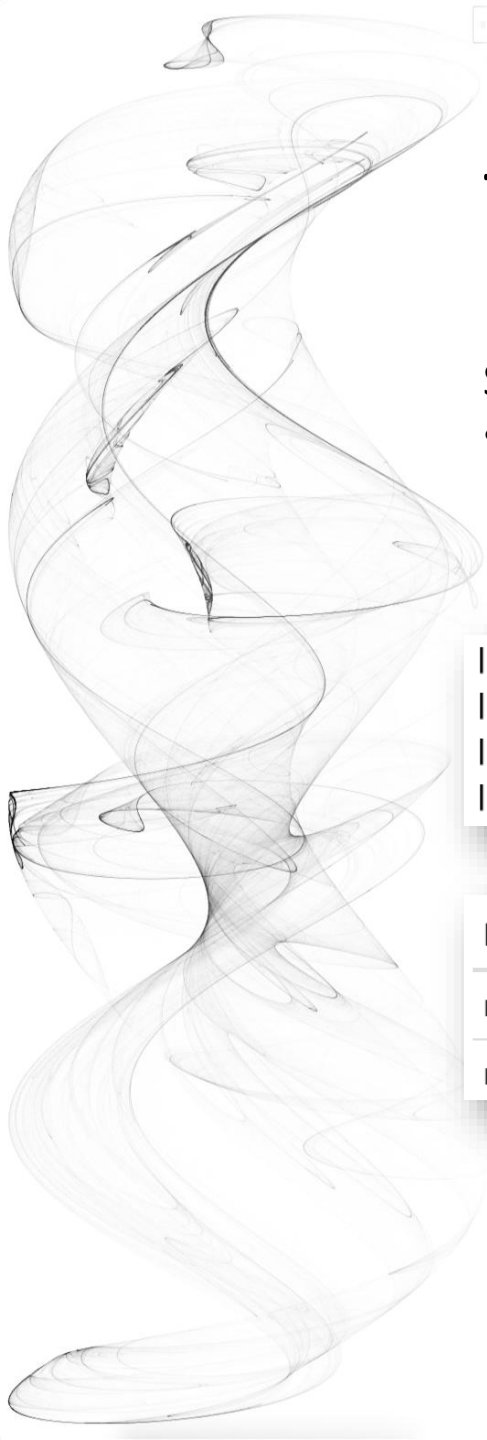
Syntax:

- Headers: first row – column name & second row – filled by dashes
- Contents: third (and all following) rows
- Columns: separated by the pipe symbol (|)

```
91 ▾ ##### Example Table of Data Structures in R
92
93 | **Structure** | **Description** | **Example** |
94 | ----- | ----- | ----- |
95 | **Vector** | A sequence of data elements of the same basic type. | `c(1, 2, 3)` |
96 | **Matrix** | A 2D array where all elements are of the same type. | `matrix(1:6, nrow = 2, ncol = 3)` |
97 | **List** | A collection of elements that can be of different types. | `list(a = 1, b = "text", c = TRUE)` |
98 | **Data Frame** | A table or two-dimensional array-like structure. | `data.frame(x = 1:3, y = c("a", "b", "c"))` |
```

Example Table of Data Structures in R

| Structure | Description | Example |
|-------------------|--|--|
| Vector | A sequence of data elements of the same basic type. | <code>c(1, 2, 3)</code> |
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| Data Frame | A table or two-dimensional array-like structure. | <code>data.frame(x = 1:3, y = c("a", "b", "c"))</code> |



Tables

Syntax:

- Headers:
 - second row – filled by dashes, use colons (:) to indicate alignment

| | | | |
|--------------|----------------|---------------|--|
| Left-aligned | Center-aligned | Right-aligned | |
| :----- | :-----: | -----: | |
| row1 | row1 | row1 | |
| row2 | row2 | row2 | |

| Left-aligned | Center-aligned | Right-aligned |
|--------------|----------------|---------------|
| row1 | row1 | row1 |
| row2 | row2 | row2 |

This abstract black and white artwork features a complex, swirling, and layered pattern. The design is composed of numerous overlapping, fluid lines that create a sense of motion and depth, reminiscent of smoke or liquid in motion. The lines are primarily black, with varying opacities that allow some to appear more prominent than others. The overall shape is elongated and vertical, with a small, dark, teardrop-like element at the top center. The background is a solid, light gray, which provides a stark contrast to the dark, intricate lines. The composition is dynamic and non-representational, focusing on the interplay of form and movement.

```
# header 1  


# > header 1 </h1> --- Line 1<br>Line 2 ---  --- <b>Bold Text</b> and **Bold Text** <i>Italic Text</i> and *Italic Text* ---


```

| Left-aligned | Center-aligned | Right-aligned |
|----------------------------|----------------|---------------|
| :----- | :-----: | -----: |
| row1 | row1 | row1 |
| row2 | row2 | row2 |

| Left-aligned | Center-aligned | Right-aligned |
|--------------|----------------|---------------|
| row1 | row1 | row1 |
| row2 | row2 | row2 |



In this lecture

- Markdown language & R markdown
- Markdown Basics
 - Components: titles, bulleted list, numbered list, checklist, link, image, code block, table
 - Formatting: italic, bold, quote, in-line code

A TINY CHEATSHEET ON



MARKDOWN

Markdown is a lightweight markup language. Its design allows it to be converted to many output formats like HTML

Titles

Heading 1

Heading 2

Heading 3

Heading 4

Heading 5

Heading 6

Text

Italic

Also Italic

****Bold****

Also Bold

Lists

- * Item 1

- * Item 2

- This works

- Too

1. Ordered

2. Lists

3. Are Fun

Links & Images

![Image Title](Image URL)

[Click Me](Link URL)

[profile](github.com/godcrampy)

Quotes

Linus Said:

```
> Cheat Sheets are Cool!  
And they are Funky
```

Checklist

- [x] this is a complete item

- [] this is an incomplete item

Code

```
```javascript  
function foo() {
 // Comment
 let a = 5;
 let b = 7;
 return a + b;
}
```  
`<inline-code>`
```